Machine Learning

Logistic Regression- Week 6

Objectives

- Recap: Linear Regression for multiple features/Matrix Formulation
- Intro to Logistic Regression
- Reading Assignment (Andrew Ng , CS229 Lecture notes)

Slide credits:

Andrew Ng, CS229 Lecture notes

Where to find me?

Office:

Science Hall 250

Office Hours:

Weds and Thurs 11:00 AM - 1:00 PM EST or by appointment

What is X?

House sizes:

2104

1416

1534

852

Predicting price of houses based on house sizes. Let's assume, our regression line is given by the following equation

$$f(x) = -40 + 0.25\,x$$

Here x in this equation would be our house sizes, just a single feature based on which we are calculating ŷ

We can write training samples in the following way:

$$\underset{\text{4x2 matrix}}{\begin{bmatrix} 1 & 2104 \\ 1 & 1416 \\ 1 & 1534 \\ 1 & 852 \end{bmatrix}} \boxed{*} \underset{\text{2x1 vector}}{\begin{bmatrix} -40 \\ 0.25 \end{bmatrix}} = \underset{\text{Predicted value (4x1)}}{\begin{bmatrix} -40 + 0.25 * 2104 \\ -40 + 0.25 * 1416 \\ -40 + 0.25 * 1534 \\ -40 + 0.25 * 852 \end{bmatrix}}$$

Matrix multiplication

What is X?

House sizes:
2104
1416
1534
852

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Here x in this equation would be our house sizes, just a single feature based on which we are calculating ŷ

We can write training samples in the following way:

4x2 matrix

2x1 vector

Predicted value (4x1)

$$\begin{bmatrix} 1 & 2104 \\ 1 & 1416 \\ 1 & 1534 \\ 1 & 852 \end{bmatrix} * \begin{bmatrix} -40 \\ 0.25 \end{bmatrix} = \begin{bmatrix} -40 + 0.25 * 2104 \\ -40 + 0.25 * 1416 \\ -40 + 0.25 * 1534 \\ -40 + 0.25 * 852 \end{bmatrix}$$

Prediction ŷ for training sample 1

Prediction ŷ for training sample 2

Prediction ŷ for training sample 4

b

w

Added to account for b

feature x, i.e. house sizes

What is X?

House sizes:	# of bedrooms
2104	3
1416	2
1534	3
852	1

Predicting price of houses based on house sizes. Let's imagine, our regression line is given by the following equation

$$f(x) = -40 + 0.25\,x^1 + 0.3x^2$$

where x^1 → house size
x^2 → # of bedrooms

We can write training samples in the following way:

feature x^1, i.e., house sizes

feature x^2, i.e., # of bedrooms

$$\begin{bmatrix} 1 & 2104 & 3 \\ 1 & 1416 & 2 \\ 1 & 1534 & 3 \\ 1 & 852 & 1 \end{bmatrix} * \begin{bmatrix} -40 \\ 0.25 \\ 0.3 \end{bmatrix} = \begin{bmatrix} -40 + 0.25 * 2104 + 0.3 * 3 \\ -40 + 0.25 * 1416 + 0.3 * 2 \\ -40 + 0.25 * 1534 + 0.3 * 3 \\ -40 + 0.25 * 852 + 0.3 * 1 \end{bmatrix}$$

4x3 matrix

3x1 vector

Predicted value (4x1)

Prediction $\hat{y}^1$

Prediction $\hat{y}^4$

In general, linear regression equation, can be written as : $w^T X + b$

Linear Regression (in Matrix Formulation)

In general, linear regression equation, can be written as : $\boxed{w^T}X + b$

Matrix

Where w^T → transpose of weight matrix
X → matrix of features

From now on, when we use X, think of it as a matrix of features, not a single variable.

[Note: review linear algebra from here and here]

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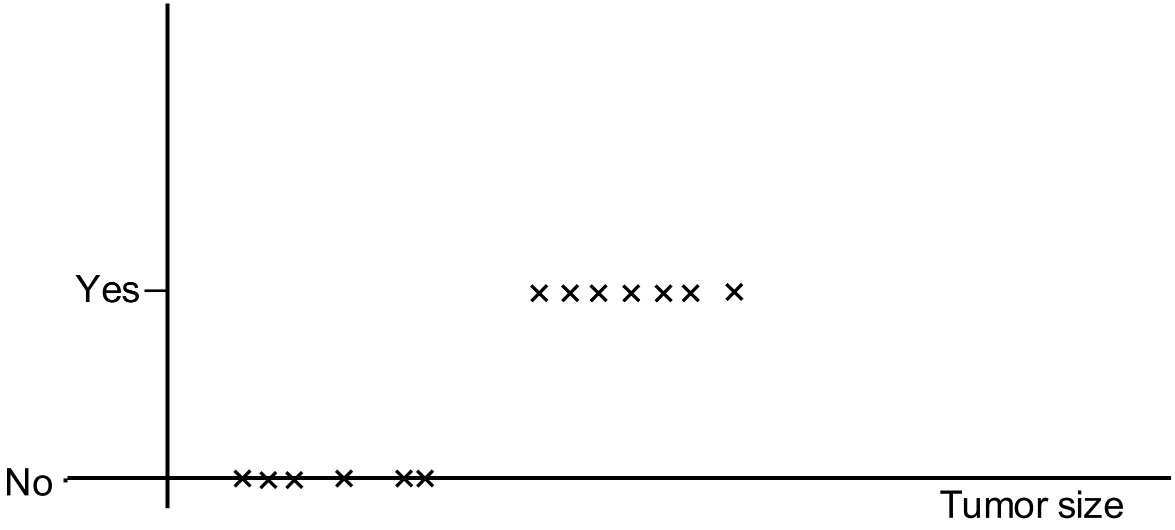
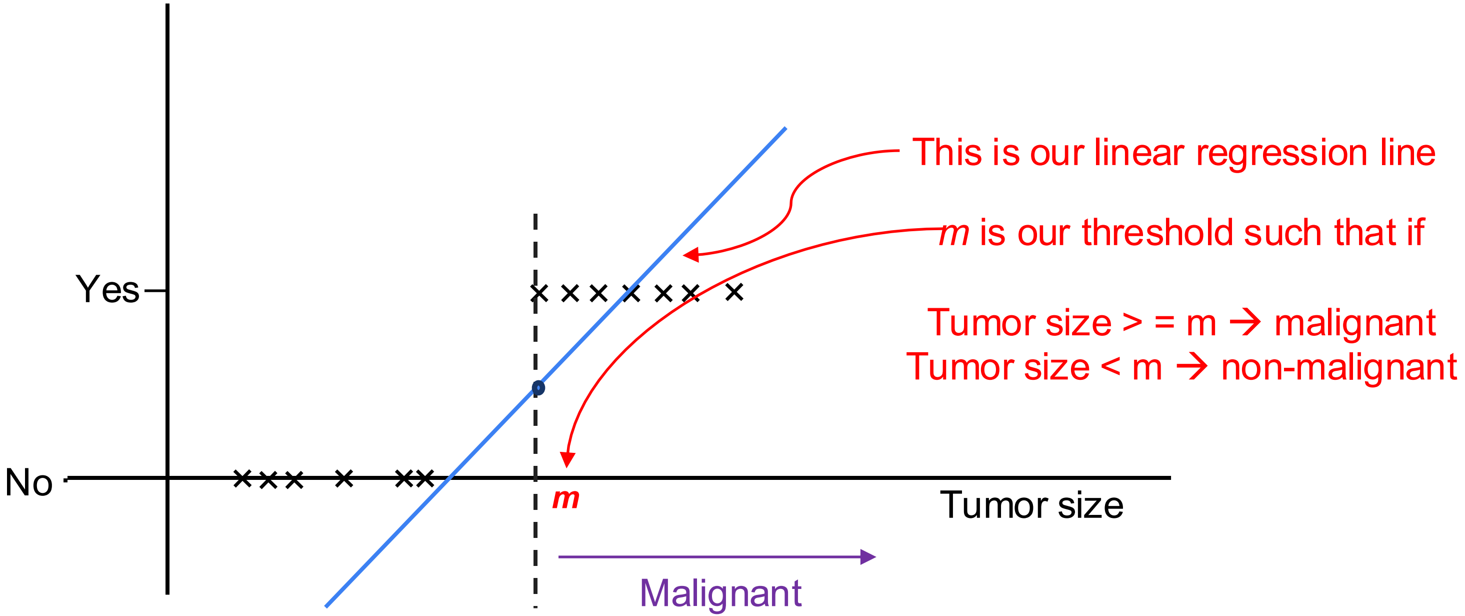
Intro to Logistic Regression

Now let's investigate a different problem. Given some tumor sizes, we want to find out if a tumor is malignant or not. Can we use linear regression here?



Intro to Logistic Regression

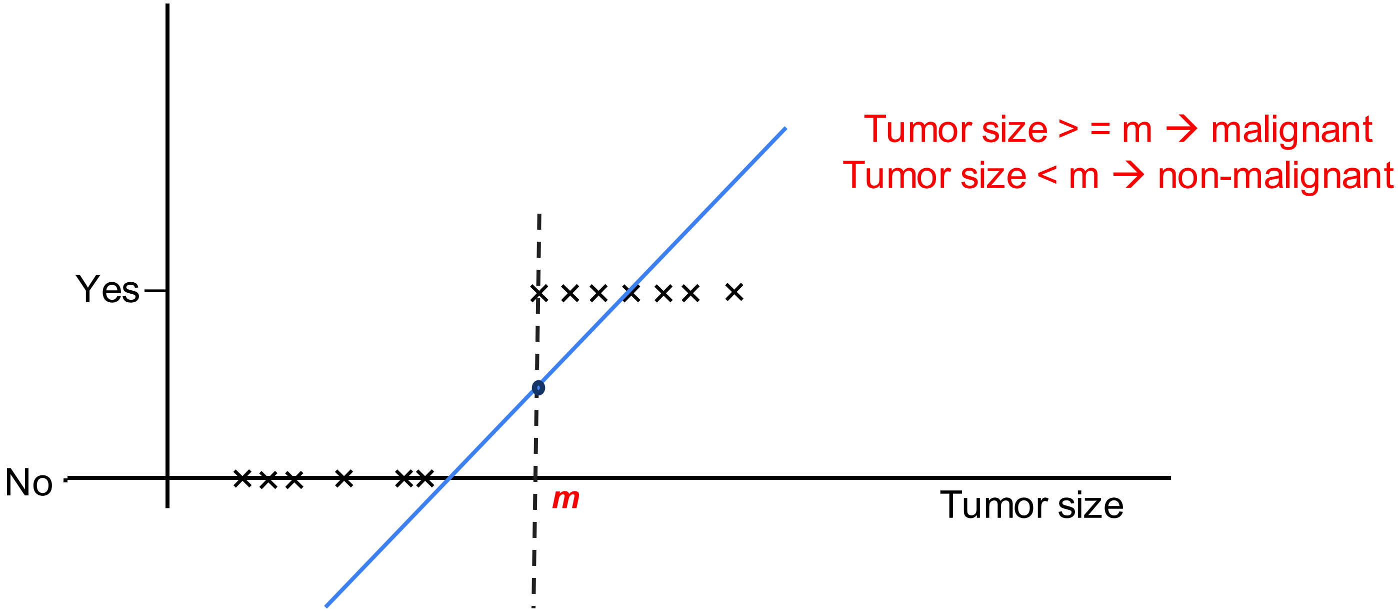
Now let's investigate a different problem. Given some tumor sizes, we want to find out if a tumor is malignant or not. Can we use linear regression here?



Issues with Linear Regression

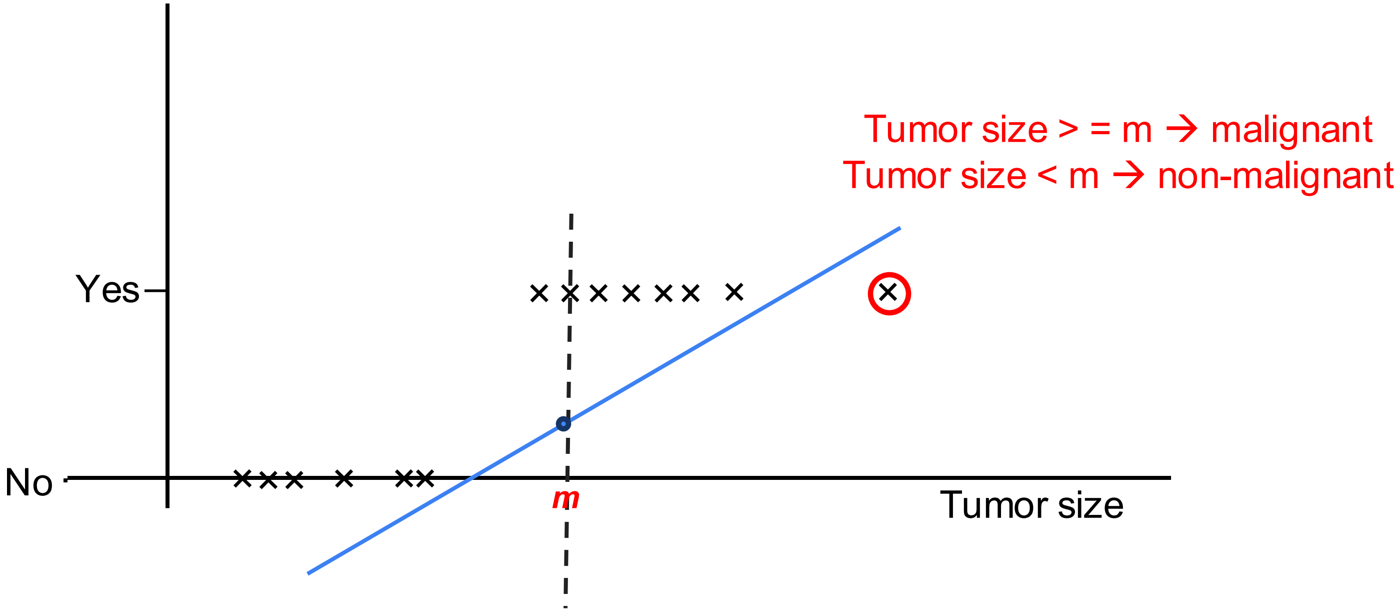
The prediction, $\hat{y}$ is a continuous value, i.e. $\boldsymbol{\hat{y} < 0}$ *or* $\boldsymbol{> 1}$

But for classification tasks such as tumor malignancy prediction, ***ŷ has to be either 0 or 1***



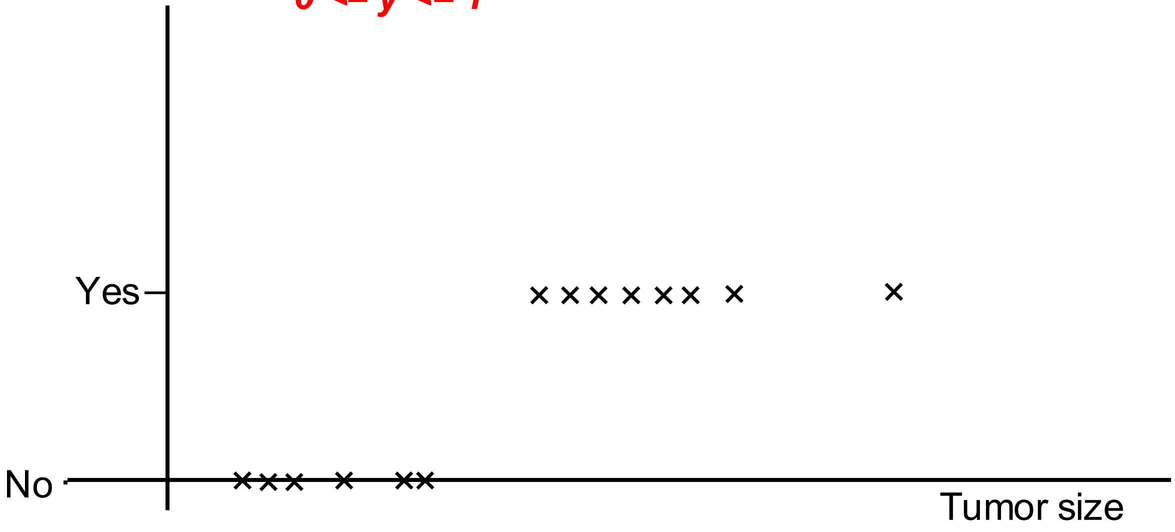
Issues with Linear Regression

Outliers (such as a new training data point) will change the threshold



Logistic Regression

We need to develop an algorithm such that
$0 <= \hat{y} <= 1$



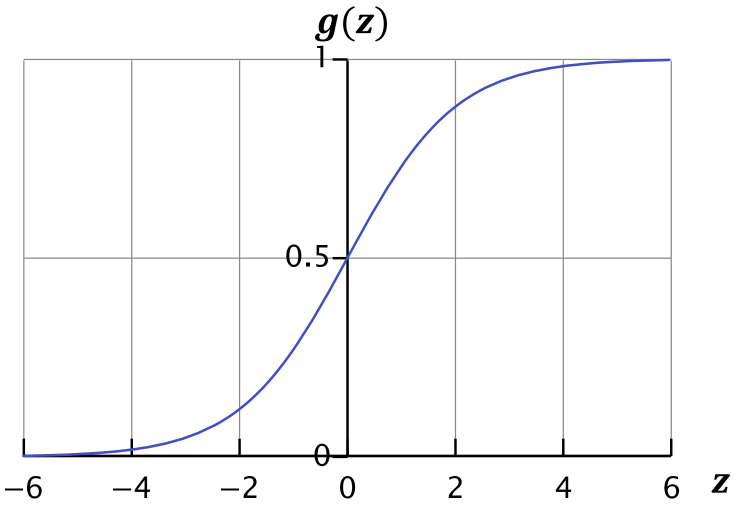
Logistic Regression

Want *0 <= ŷ <= 1,*
Linear regression equation : $f(x) = wT\,x + b$

Now we define logistic regression as $f(x) = g(wT\,x + b)$

Where $g(z) = \frac{1}{1+e^{-z}}$

$g(z)$ is called the **sigmoid** or **logistic** function



ANY QUESTIONS?